

Education Spending and Economic Growth: Short- and Long-Term Effects in a Global Panel*

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Abstract

The role of public education spending in promoting economic growth is widely debated, yet empirical evidence distinguishing short-run from long-run effects remains limited. This paper estimates these effects using a CS-ARDL model with error correction for a global panel of 107 countries (1970–2019), disaggregating spending by primary, secondary, and tertiary levels. A 1% increase in real per capita education expenditure is associated with 0.031% higher long-run output for aggregate spending, 0.019% for both primary and secondary, and 0.013% for tertiary, with significant effects in both the short and the long run. Higher educational achievement, proxied by PISA scores, amplifies the growth impact of spending. These findings underscore the role of education quality in enhancing the productivity of public investment.

Keywords: educational expenditure, economic growth, error correction model

JEL Classification: H52, I25, O10

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1 Introduction

Investment in education is widely recognized for its long-term direct effects on economic growth and productivity. The primary mechanism is through enhancing the quality of the labor force. Through the development of human capital, education provides individuals with the skills and knowledge needed to become more productive and innovative. This, in turn, fosters an environment conducive to entrepreneurship and the adoption of new technologies, both of which are key drivers of economic development.¹

Beyond its well-documented conceptual importance, education represents one of the largest categories of public expenditure worldwide. According to recent estimates, governments globally spend approximately 4.3 percent of GDP on education, a figure that has remained broadly stagnant over the past decade and lies at the lower bound of the 4–6 percent benchmark recommended by UNESCO for achieving sustainable education systems (UNESCO, 2023). This aggregate masks substantial cross-regional heterogeneity: high-income countries typically allocate between 4.5 and 5 percent of GDP to education, while countries in Sub-Saharan Africa spend closer to 3.4 percent on average, despite facing higher demographic pressures and lower baseline learning outcomes (World Bank, 2023; UNESCO, 2023).

Differences are even more pronounced when measured in per-student terms. In 2022, public education spending averaged roughly USD 8,500 per school-age child per year in high-income countries, compared to just USD 55 in low-income countries—a more than 150-fold gap in educational investment per child (UNESCO, 2024). As a result, although low- and lower-middle-income countries account for nearly half of the world’s school-age population, they represent less than 1 percent of total global education spending, while high-income countries account for over 60 percent (UNESCO, 2024). These stark disparities highlight the central role of public education spending in shaping global inequality in human capital formation and motivate a closer examination of its macroeconomic consequences.

Beyond its direct impact on productivity, education generates several indirect benefits contributing to economic stability and growth. Educated individuals are more likely to participate in the labor force and secure higher-paying jobs, which leads to increased income and reduced poverty levels. Additionally, a more educated workforce tends to experience lower job turnover and unemployment rates, as individuals possess the skills needed to adapt to changing job market demands. This eases the strain on state welfare programs and social safety nets.²

Recent research has further expanded the scope of benefits associated with increased investment in education. Beyond economic indicators, education contributes to social well-being by reducing crime rates and providing greater opportunities for

¹For instance, Hanushek and Woessmann (2010) show a strong link between educational attainment and economic growth, indicating that countries with higher levels of education tend to experience faster economic growth.

²See, for example, Oreopoulos and Salvanes (2011), who highlight that education significantly improves labor market outcomes, including employment prospects and earnings.

women to participate in the workforce. Educated women are more likely to delay childbirth and pursue careers, leading to lower rates of teenage pregnancies and improved outcomes for children. [Lochner and Moretti \(2004\)](#) found that higher levels of education are associated with significant reductions in crime rates, suggesting that education plays a crucial role in promoting social stability. Moreover, [Psacharopoulos and Patrinos \(2018\)](#) provide evidence that education leads to better health outcomes, including lower fertility rates and a reduced incidence of teenage pregnancies.

Although the benefits of education investment are well-documented, they often take time to fully materialize. Human capital development and its subsequent economic impact are gradual processes. However, the scale of education investment also implies that certain short-term economic effects can be observed. Increased educational spending can stimulate economic activity by creating jobs, boosting demand for educational materials and services, and enhancing infrastructure. The multiplier effect of such spending can lead to immediate boosts in economic activity, even as the long-term benefits continue to unfold.³

This paper examines whether increases in education expenditure generate measurable effects on the economic growth of countries, both in the short and long run. Using a dynamic panel approach that distinguishes between immediate and long-term responses, we estimate the average returns on education investment globally and across disaggregated educational levels, namely primary, secondary, and tertiary. Rather than comparing regional aggregates, we focus on identifying sources of heterogeneity in the growth impact of education spending. In particular, we emphasize how country-specific characteristics, possibly linked to the quality of expenditure, especially learning outcomes as proxied by standardized test scores (e.g., PISA scores), can amplify these effects.

The main objective of this paper is to estimate the short- and long-run effects of public education expenditure on economic growth in a global panel and to assess how these effects vary by education level and by education quality. We test the hypothesis that education expenditure has a positive impact on output in both the short and long run, and that this impact is heterogeneous across levels of education and across countries with different levels of educational achievement (proxied by PISA scores).

Our contribution addresses three gaps in the literature reviewed in Section 2. First, we provide evidence from a global panel that distinguishes short-run from long-run effects of education spending, using a cross-sectionally augmented ARDL model with an error correction representation (CS-ARDL-ECM) and a sample of 107 countries (1970–2019), a setting in which such dynamic effects remain underexplored. Second, we examine how education quality (proxied by PISA scores) conditions the growth impact of spending, a dimension that prior global studies have rarely incorporated. Third, we disaggregate public education expenditure by level (primary, secondary, and tertiary), allowing us to compare the growth returns of different types of investment and to inform where spending is most effective.

³Using data for the United States, [De Ridder et al \(2020\)](#) and [Angelopoulos et al \(2008\)](#) show that local governments prioritizing education spending often see immediate and sustained economic benefits.

The CS-ARDL-ECM methodology has been extensively used in growth-related literature to understand the relationship with variables that are closely aligned with GDP over time. Broadly, several papers have studied the relationship between various measures of human capital and economic growth (e.g., Afzal et al (2010); Adeyemi and Ogunsola (2016); Mabrouki (2023); Gebrehiwo (2016)).

Other research more closely aligned with our paper specifically examines the role of public spending on education and its relationship with GDP in various regions of the world (e.g., Coman et al (2023); Ramli et al (2022); Mohamed et al (2021); Ifa and Guetat (2018); Kiran (2014); Mallick et al (2016)). Regarding the relationship between education spending and growth in Latin America and the Caribbean, Kiran (2014) studied the effect of education expenditure on economic growth for 18 countries in the region from 1970 to 2009. This study uses a VECM methodology that identifies structural breaks to capture the potential impact of educational policies that modify the cointegration relationship for each country. The findings reveal cointegration between education spending and economic growth for the countries in Latin America and the Caribbean, with the exception of Chile, Guyana, Jamaica, Nicaragua, Paraguay, Peru, and Uruguay. Another study closely related to our research question was conducted by Mallick et al (2016). It examines the relationship between GDP and education spending across 14 Asian countries over the period of 1973-2012. Using a panel VECM methodology, the authors find positive and significant effects of education spending on growth. They also apply Granger causality tests to formalize the unidirectional effect of this relationship (i.e., to test whether education spending drives economic growth rather than the opposite).

We find that a 1 percent increase in overall education expenditure is associated with a 0.031 percent increase in long-run output. Elasticity estimates by level of education are lower, ranging between 0.013 percent and 0.019 percent. Importantly, we also observe that these elasticities vary considerably and are strongly associated with student achievement, as measured by PISA scores. Specifically, countries with higher average PISA scores tend to exhibit a greater elasticity of output with respect to education expenditure. In particular, countries with PISA scores near the average for advanced economies, approximately 30 percent higher than those of an average emerging or developing country, show long-run elasticities that are nearly four times higher than those of countries with lower PISA scores. This suggests the presence of nonlinearities in the distribution of elasticities. While it is true that countries with higher education spending tend to exhibit higher average PISA scores, this may also reflect differences in the quality of spending. In this regard, higher PISA performance may be indicative of more effective, higher-quality expenditure, which in turn is associated with higher long-run elasticities of output with respect to per capita education spending.

Finally, our results suggest that increasing education expenditure also yields short-run impacts, possibly through demand-side channels. These short-term effects appear relatively consistent across different levels of education quality. However, long-term impacts vary significantly depending on the quality of education spending.

The rest of the paper is organized as follows. Section 2 reviews the literature. Section 3 presents the methodology, including the data and the empirical strategy. Section 4 reports the results of estimating the CS-ARDL models using measures of expenditure disaggregated by level of education. Section 5 examines how education expenditure quality, as proxied by the PISA scores of each country, conditions these impacts. Section 6 uses the estimation of the previous sections to illustrate the short and long-run adjustment dynamics of GDP per capita in response to an increase in education expenditure. Section 7 discusses the findings, their interpretation, policy implications, and limitations. Section 8 concludes.

2 Literature Review

The link between education and economic growth is well-established in economic theory. In endogenous growth models, human capital accumulation plays a central role in sustaining long-run growth (Lucas, 1988). Education increases the productivity of labor and can generate positive externalities that fuel technological progress. Mankiw et al (1992) showed that including human capital (schooling) in the neoclassical growth model helps explain income differences across countries, aligning with earlier findings that nations with higher educational attainment tend to grow faster (Barro, 1991). These foundational works provided a theoretical rationale for public investment in education as a catalyst for growth. In particular, models such as Lucas (1988) imply that government spending on education, by fostering human capital, should raise an economy's steady-state growth rate. Similarly, Barro (1990) and others argued that productive public expenditures (like education) can augment growth, although the growth payoff may depend on financing conditions (taxation) and diminishing returns.

Empirical evidence on the education-growth nexus has generally found a positive relationship when education is measured by enrollment or attainment. However, studies focusing specifically on public education *expenditure* have yielded mixed results, especially in earlier cross-country analyses. Some cross-country regressions in the 1980s and 1990s found little or no robust effect of education spending on growth once other factors were controlled. For instance, Landau (1986) and Levine and Renelt (1992) reported that government education expenditures were not a statistically robust determinant of growth in broad samples of countries. Similarly, Easterly and Rebelo (1993) found positive growth effects of public education investment only under certain model specifications, and Devarajan et al (1996) concluded that education spending had an insignificant (or even negative) impact on growth in developing countries. These ambiguous results contrast with the clear predictions of theory and raised questions about whether simply spending more on education guarantees better growth outcomes.

Researchers noted several potential explanations: the productivity of spending may vary across countries; the effect of spending could be non-linear or contingent on how it is financed; and increases in spending do not always translate into actual improvements in human capital due to inefficiencies or poor quality of education (Blankenau and Simpson, 2004; Pritchett, 2001). In fact, Blankenau and Simpson (2004) demonstrate theoretically that the growth impact of education expenditure can be offset by distortionary taxes required to finance it, implying a non-monotonic relationship where spending boosts growth up to a point and then may hinder it if taxation costs become

too high. This insight helps reconcile why simple cross-country regressions sometimes failed to find a positive effect: without accounting for the efficiency of spending and the funding mechanisms, the net growth impact of public education expenditure can appear statistically elusive.

In the 2000s and 2010s, a large number of empirical studies turned to country-specific time-series and regional panel analyses to better capture the education expenditure–growth relationship. Many of these works applied cointegration techniques and autoregressive distributed lag (ARDL) models to distinguish short-run versus long-run effects. The general finding is that public education spending does contribute positively to long-run economic growth in many contexts, even if short-run effects are modest. For example, [Afzal et al \(2010\)](#) examined the case of Pakistan and found evidence of a cointegrating relationship between education expenditures and GDP, with education spending exerting a significant positive impact on long-run growth (though short-run effects were smaller). Similar ARDL-bound tests in other developing economies echo this result: [Adeyemi and Ogunsola \(2016\)](#) report that human capital expenditures in Nigeria have a positive and significant long-run effect on GDP, and [Gebrehiwo \(2016\)](#) find a growth-promoting impact of education spending in Ethiopia in the long term. In North Africa, [Ifa and Guetat \(2018\)](#) show that increasing public education expenditure led to higher per capita GDP in Tunisia and Morocco over time, using an ARDL approach, and [Ramli et al \(2022\)](#) report a positive long-run elasticity of output with respect to education investment in Algeria. Likewise, for Egypt, [Mohamed et al \(2021\)](#) find that human capital expenditure improves economic growth in the long run.

These country-level studies consistently indicate that while education spending may not immediately translate into growth (often the short-run coefficients are insignificant or smaller in magnitude), there is a meaningful long-run payoff. The error-correction terms in these models are significant, confirming that GDP adjusts towards a higher equilibrium path after increases in education expenditure.

Beyond individual countries, researchers have also investigated this relationship in multi-country panels, often focusing on specific regions. These studies typically allow for heterogeneous country dynamics while identifying an average effect. For instance, [Kiran \(2014\)](#) analyzes 18 Latin American countries from 1970–2009 using a panel vector error-correction model (VECM) that accounts for potential structural breaks. The study finds cointegration between public education spending and output in most countries (with a few exceptions) and reports a positive long-run effect of spending on growth. In addition, [Kiran \(2014\)](#) finds evidence of Granger causality running from education expenditure to growth, supporting the view that investment in education is a driver of economic performance rather than merely a response to higher income. Focusing on Asia, [Mallick et al \(2016\)](#) investigate 14 major Asian economies over 1973–2012 and likewise conclude that education expenditure has a significant positive impact on long-run growth. Their panel VECM estimates indicate that higher education spending leads to higher steady-state GDP, and Granger causality tests in that study confirm a unidirectional causality from education spending to growth. More recently, [Coman et al \(2023\)](#) examine countries in Central and Eastern Europe

and find that public education spending boosts economic growth in the long run. Using an ARDL approach with structural break adjustments, they show that even when accounting for policy regime changes, the positive linkage between education expenditure and per capita GDP holds.

Another notable study is by [Mabrouki \(2023\)](#), who employs a dynamic common-correlated effects panel (CS-ARDL) model for Scandinavian countries. They report that human capital measures (including education spending and innovation proxies like patents) have a significant long-run contribution to growth, reinforcing the general consensus that education-related investments are growth-enhancing over the long horizon. Across these regional studies, a common theme is that short-run effects of education spending on output are often weaker or transient, whereas the long-run effects are robustly positive. This aligns with the intuition that the benefits of education, such as higher productivity, a more skilled labor force, and technological innovation, accrue over time.

While the literature broadly supports a positive long-term impact of education spending on growth, it has also highlighted several complexities and nuances. One such insight is the importance of education quality and efficiency. Simply spending more on education does not guarantee better outcomes if the funds are not used effectively. [Pritchett \(2001\)](#) famously asked “Where has all the education gone?”, showing that many countries with rapid expansions in schooling did not experience commensurate growth, likely due to low education quality and weak institutions. In a related vein, [Hanushek and Kimko \(2000\)](#) and subsequent work by [Hanushek and Woessmann \(2010\)](#) demonstrate that the cognitive skills of the population, measured by international test scores, have a powerful effect on economic growth. These studies find that once you account for educational quality, the role of mere school attainment or spending becomes much clearer: nations that produce higher test scores (indicating better quality education) enjoy significantly higher growth rates than those with lower scores, even if spending levels or years of schooling are similar. This suggests that the growth payoff to education spending is strongly mediated by the efficiency of spending in producing human capital.

However, most empirical studies on education expenditure and growth have not explicitly included quality measures like test scores or learning outcomes in their models. As a result, a gap remains in understanding how improvements in education quality might amplify (or condition) the effect of spending on growth. The present research addresses this by incorporating an interaction of public spending with educational achievement (PISA scores) to assess whether “quality-adjusted” education spending yields higher growth dividends, a question that previous global studies have scarcely explored.

Another notable gap in the existing literature is the relative dearth of evidence from global panel analyses that distinguish between short-run and long-run effects of education spending. Many early cross-country studies essentially captured long-run associations (e.g., using decade-averaged data or steady-state growth rates) and could not separate immediate cyclical impacts from enduring growth effects. Conversely,

most time-series studies focus on individual countries, and regional panels cover subsets of countries with similar characteristics. A comprehensive global panel study can provide insights into the generalizability of results across diverse economies, as well as greater statistical power to detect long-run relationships. Moreover, a global panel with a suitable econometric approach allows one to account for heterogeneity and cross-country spillovers. To our knowledge, few works have applied modern dynamic panel techniques (such as error-correction models that yield short- and long-run estimates) to a truly global sample. [Baldacci et al \(2008\)](#) is a partial exception, examining 118 developing countries and finding that increases in education (and health) expenditures indirectly promote growth by improving human capital outcomes, but even that study did not explicitly report short-run vs. long-run elasticities in an integrated framework.

The present study contributes to filling this gap by using a panel of over 100 countries worldwide and by employing a cross-sectionally augmented ARDL approach that jointly estimates short- and long-term effects while addressing cross-sectional dependence. This approach builds on recent advances in panel time-series econometrics (e.g., [Chudik et al, 2016](#); [Ditzen, 2021](#)), and allows us to test whether the long-run equilibrium relationship between education spending and GDP differs from the short-run dynamics, a nuance that earlier global analyses could not capture due to methodological limitations.

A further limitation in much of the existing research is the treatment of education expenditure as an aggregate, without disaggregating by education level. Government spending on primary education might have a different economic impact than spending on secondary or tertiary education. From a human capital perspective, primary and secondary education create fundamental literacy and skills that broaden the productive base, whereas tertiary education may foster advanced skills and innovation but directly affects a smaller portion of the workforce. Microeconomic evidence on rates of return has often found diminishing returns to higher levels of education in developing countries, with primary education yielding the highest social returns at the margin ([Psacharopoulos and Patrinos, 2018](#)). This raises the question of whether public expenditure is most growth-enhancing when directed toward basic education or toward higher education.

However, relatively few macro-level studies have attempted to distinguish the growth effects of education spending by level. An exception is the work of [Mabrouki \(2023\)](#) (focused on Scandinavia), which included disaggregated spending and suggested that different education levels can have quantitatively different effects. Some country studies also hint at level-specific impacts: for example, research on India’s states by [Self and Grabowski \(2004\)](#) found that primary education enrollment contributed more to state economic growth than tertiary enrollment. Overall, the lack of disaggregation in most cross-country analyses means we know little about whether investing in primary education versus secondary or tertiary education has a more potent effect on growth in various contexts. The present study addresses this gap by explicitly separating public education expenditures into primary, secondary, and tertiary and estimating their individual short-run and long-run growth effects in a unified model. This disaggregation can reveal important heterogeneity; for instance, it allows us to test whether

primary education investment offers the quickest growth dividends in the short run, or whether tertiary education, despite a smaller immediate impact, accumulates to substantial long-run gains.

In summary, the literature establishes education as a key driver of economic growth, and an increasing number of empirical studies confirm that public education spending, in general, supports higher long-run growth. But the findings also underscore that the impact of spending is not automatic—it depends on factors like the efficiency/quality of education and the allocation of spending across educational stages. Important research gaps remain. First, there is a lack of global panel studies that simultaneously examine short- vs. long-run effects of education spending; most prior work is either cross-sectional or country-specific, making it difficult to draw general conclusions about dynamic effects across countries. Second, there has been limited exploration of how education quality interacts with spending in influencing growth; few studies incorporate direct measures of learning outcomes or quality, despite evidence that quality is crucial. Third, there is a need for disaggregated analysis by education level; the growth contributions of primary, secondary, and tertiary education spending may differ, yet most empirical analyses use only aggregate spending data. These gaps in the literature motivate our study. In the analysis that follows, we contribute to the literature by using a global panel of 107 countries over five decades to assess short- and long-term effects of public education expenditure on growth, while accounting for education quality and disaggregating spending across primary, secondary, and tertiary levels. By addressing these dimensions, we aim to provide a more nuanced understanding of how and under what conditions public education spending translates into economic growth.

3 Methodology

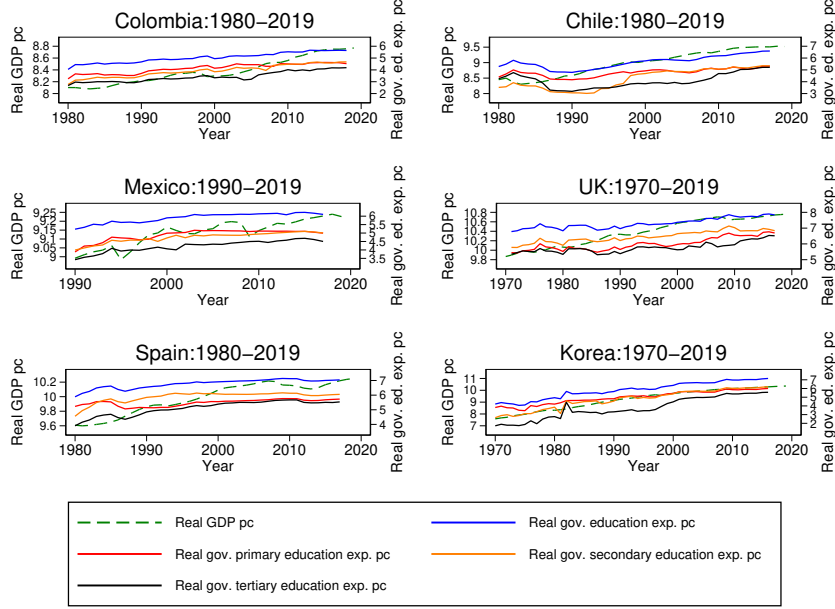
3.1 Data

To contribute to the literature on the relationship between educational expenditure and economic outputs, we analyze the short- and long-term co-movements between real GDP per capita and different measures of expenditure in education. The dependent variable is real GDP per capita, the standard measure of aggregate output in the growth literature. On the expenditure side, we use government education spending disaggregated by level (primary, secondary, and tertiary), sourced from UNESCO's UIS database—the most comprehensive and internationally comparable dataset on public education finance. All variables are expressed in real per capita terms to ensure consistency between dependent and independent variables and to avoid spurious correlations driven by inflation or population size. The sample comprises 107 countries, determined by the intersection of GDP data from the World Bank and education expenditure data from UNESCO/UIS over a sufficient time span. No additional selection criteria are imposed, yielding broad coverage across income levels and geographic regions.⁴ To illustrate this relationship, we first plot these data for a series of selected

⁴See Appendix [Table A1](#) for a description of the data and their sources.

countries in [Figure 1](#). This pattern, repeated across many countries, shows a potentially close relationship between the educational expenditure in real per capita terms for different educational levels and real GDP per capita.

Fig. 1: Logarithmic series of GDP, educational expenditure, primary education expenditure, secondary education expenditure, and tertiary education expenditure by country



Note: This figure plots real GDP per capita and real per capita government education expenditures by education level for a selected set of countries from 1970 to 2019. All series are shown in natural logarithms. Data are sourced from the World Bank's World Development Indicators (WDI) and UNESCO's UIS database.

An analysis of [Figure 1](#) reveals a potentially parallel trend between aggregate output and aggregate educational expenditure. A similar pattern emerges when disaggregating expenditure by education level. Moreover, this relationship does not seem to differ when comparing developed and developing countries over a similar period.⁵ Appendix [Table D4](#) shows the pairwise correlation of the critical variables in this study. In line with the patterns observed in [Figure 1](#), the results show a strong correlation across time and minimal differences between advanced and emerging economies.

3.2 Empirical strategy

To conduct this analysis, we use the flexibility of ARDL/ECM models. An Autoregressive Distributed Lag model (ARDL) is an econometric model that allows for the

⁵Details on the sample of countries can be found in Appendix [Table B2](#) and basic descriptive statistics in Appendix [Table C3](#).

analysis of a potential long-term relationship between sets of variables, not necessarily all integrated of order 1. This was initially developed in Pesaran et al (1995, 2001). One of the main features of the ARDL model is that it combines both autoregressive terms and distributed lags. In this paper, we exploit its error correction (ECM) representation, which captures both short-term and long-term dynamics in a single model specification.

The key question in this study is to explore the magnitude of these correlations and analyze the differences between long- and short-term co-movements. To do this, we employ a CS-ARDL(p,q) specification (cross-section ARDL generalized for panel study). The CS-ARDL(p,q) model can be summarized in the following specification:

$$Y_{it} = c_i + \sum_{j=1}^{p_i} \phi_{ij} Y_{i,t-j} + \sum_{j=0}^{q_i} \beta'_{ij} \mathbf{x}_{i,t-j} + u_{i,t} \quad (1)$$

In this specification, i indexes countries, t indexes time, and j indexes lags. For each country i , $Y_{i,t}$ denotes the real GDP per capita and $\mathbf{x}_{i,t}$ is a $K \times 1$ vector of real per capita government education expenditures. The lag orders $p \geq 1$ and $q \geq 0$ determine the number of autoregressive and distributed lag terms, respectively; for simplicity, we assume that q is the same for all variables in $\mathbf{x}_{i,t}$. Although the general CS-ARDL framework allows for country-specific lag orders, in our implementation p and q are common across countries and selected for the full panel using the Bayesian information criterion (BIC), which balances goodness of fit and parsimony. The slope coefficients, however, remain fully heterogeneous across countries. Specifically, we are interested in the error correction representation, which takes the following form:

$$\Delta Y_{i,t} = c_i - \alpha_i (Y_{i,t-1} - \boldsymbol{\theta}'_i \mathbf{x}_{i,t}) + \sum_{j=1}^{p_i-1} \psi_{yi,j} \Delta Y_{i,t-j} + \sum_{j=0}^{q_i-1} \psi'_{xi,j} \Delta \mathbf{x}_{i,t-j} + u_{i,t}. \quad (2)$$

In the ECM representation, the short-run coefficients $\psi_{yi,j}$ and $\psi_{xi,j}$ are reparameterized functions of the original ARDL parameters ϕ_{ij} and β_{ij} .⁶ The adjustment speed coefficient is given by $\alpha_i = 1 - \sum_{j=1}^{p_i} \phi_{ij}$ and the long-term coefficients take the form $\boldsymbol{\theta}_i = \frac{\sum_{j=0}^{q_i} \beta_{ij}}{\alpha_i}$. The estimation of this representation allows us to simultaneously capture the short- and long-term parameters for each country, as well as the adjustment speed. Using the methodology developed in Chudik et al (2016), we can retrieve the average of the parameters of interest (in particular the average $\boldsymbol{\theta}_i$) for the entire panel and draw inferences about their significance using the delta method. The CS-ARDL estimator includes cross-sectional averages of the regressors to control for cross-sectional dependence (e.g., common global shocks), which is important in panels of many countries.

⁶Note that the equilibrium correction term uses contemporaneous $\mathbf{x}_{i,t}$ rather than $\mathbf{x}_{i,t-1}$; this follows from the inclusion of the contemporaneous regressor ($j = 0$) in equation (1), as in the parameterization of Chudik et al (2016).

A key requirement in this methodology is a sufficiently long time series combined with a broad worldwide panel, both conditions satisfied in our data. Our dataset covers 107 countries (35 advanced and 72 emerging and developing) with periods spanning from 1970 to 2019, depending on the country. This allows us to illustrate and compare countries with significantly different educational and GDP per capita trajectories.

Another critical requirement to accurately estimate a CS-ARDL model is for the variables studied to be cointegrated. In turn, a key condition to analyze cointegration is for the variables to be integrated of the same order (preferably of the first order). To test this, we implement the panel unit root test developed by Im et al (2003),⁷ which is applicable to unbalanced panels. The null hypothesis is that all panels contain a unit root, against the alternative that at least some panels are stationary. The results for the series of interest are summarized in Table 1.

Table 1: IPS Panel Unit Root Test

	Z	p-value
<i>Whole Sample (107)</i>		
Real GDP pc (log)	-0.63	0.26
Real gov. education exp. pc (log)	-1.45	0.07
Real gov. primary education exp. pc (log)	-2.37	0.01
Real gov. secondary education exp. pc (log)	-2.13	0.02
Real gov. tertiary education exp. pc (log)	-0.78	0.22
Δ Real GDP pc (log)	-41.80	0.00
Δ Real gov. education exp. pc (log)	-34.89	0.00
Δ Real gov. primary education exp. pc (log)	-35.55	0.00
Δ Real gov. secondary education exp. pc (log)	-36.04	0.00
Δ Real gov. tertiary education exp. pc (log)	-34.19	0.00

Note: The Im-Pesaran-Shin (IPS) test for panel unit roots relaxes the assumption of a common autoregressive coefficient across panels. The null hypothesis states that all panels contain a unit root, while the alternative allows a nonzero fraction of panels to be stationary. This approach accommodates heterogeneity in dynamics across countries or units. See Im et al (2003).

The analysis of Table 1 above indicates that there is evidence that most variables in the dataset are integrated of the first order (I(1)), that is, they are stationary in first differences. In summary, this information allows us to conduct the desired cointegration analysis and proceed with the estimation of the CS-ARDL model.

The results summarized in Table 2 provide strong evidence of cointegration between real GDP per capita and real per capita government expenditure on education across most specifications. We observe statistically significant cointegration with total and secondary education expenditures, while the results for primary and especially tertiary education expenditures are weaker or mixed. The panel cointegration results allow us

⁷Other tests in this category include those developed in Levin et al (2002), Choi (2001), and Breitung and Das (2005).

to adequately estimate the CS-ARDL models described earlier. The results of these estimations are reported in the following section.

Table 2: Kao Panel Cointegration Test Results

	Real gov. education exp. pc (log)	Real gov. primary education exp. pc (log)	Real gov. secondary education exp. pc (log)	Real gov. tertiary education exp. pc (log)
<i>Whole Sample (107)</i>				
Modified DF t	-4.46 (0.0000)	-1.40 (0.0802)	-3.65 (0.0001)	1.06 (0.1441)
Dickey-Fuller t	-3.59 (0.0002)	-1.85 (0.0325)	-3.12 (0.0009)	1.27 (0.1012)
Augmented DF t	-3.21 (0.0007)	-1.68 (0.0469)	-3.19 (0.0007)	1.15 (0.1261)
Unadjusted modified DF t	-1.44 (0.0743)	1.10 (0.1349)	-0.50 (0.3095)	3.22 (0.0007)
Unadjusted DF t	-1.98 (0.0238)	-0.16 (0.4353)	-1.33 (0.0915)	3.15 (0.0008)

Note: We test for cointegration between real GDP per capita (Y) and education expenditure variables. Each cell reports the Kao test statistic with its p-value in parentheses. The null hypothesis is that the panels are not cointegrated. The test includes individual intercepts but no trend. Similar results are obtained using the Pedroni and Westerlund cointegration tests. Results from these alternative methodologies are available upon request.

4 Investment and Growth: A Breakdown by Type of Expenditure

This section evaluates whether increases in government education expenditure are associated with measurable impacts on economic growth in both the short and long run. The analysis focuses on disaggregated education spending by level (primary, secondary, and tertiary) to identify which types of investment are most effective at driving sustained output per capita growth.

Table 3 presents the main results from a CS-ARDL model with an error correction structure, which distinguishes short-run dynamics from long-term relationships while controlling for unobserved common factors. The dependent variable is the logarithm of real GDP per capita; the regressors are per capita real education expenditure (in logs) by level. The sample includes countries with data on primary, secondary, and tertiary expenditure.

The results show that education investment is positively and significantly associated with output growth in both the short and long term, across all levels of education. Columns 2 through 4 in the table break down the elasticities by type of expenditure, while the first column presents the effect of aggregate education expenditure, combining spending at the primary, secondary, and tertiary levels.

The long-term coefficients reflect the elasticity of output per capita with respect to education spending. These estimates are statistically significant at conventional levels across all categories. A 1 percent increase in real per capita education expenditure leads to a 0.031 percent increase in long-run output for aggregate spending, 0.019

percent for both primary and secondary education, and 0.013 percent for tertiary education.⁸ These results suggest that while all levels of education contribute to long-run economic performance, aggregate spending exhibits the highest overall elasticity. Notably, the effects for primary and secondary education are slightly higher compared with those for tertiary education, though all effects are economically significant.

The elasticities should be interpreted as follows: a sustained 10 percent increase in real per capita expenditure on primary or secondary education is associated with a 0.19 percent higher level of output in the long run. While modest in absolute terms, these effects are significant given the persistent nature of education-driven human capital accumulation and its compounding impact on growth over time.

Short-term dynamics, captured by the coefficient on the first difference of education expenditure, are also positive and statistically significant, albeit smaller in magnitude: 0.024 for aggregate expenditure, 0.014 for primary, 0.015 for secondary, and 0.010 for tertiary education. These results are consistent with the idea that education spending does not immediately translate into productive output gains, given the time-intensive nature of learning and human capital formation. However, positive short-run effects suggest that some returns begin to materialize even in the early stages, potentially through channels such as employment effects, resource allocation, or institutional responses to investment.

The bottom panel reports the error correction coefficient ($-\alpha_i$), whose reported estimates are approximately -0.77 and statistically significant at the 1 percent level. The implied speed of adjustment is $\alpha_i \approx 0.77$, meaning that approximately 77 percent of the gap between current and long-run equilibrium output is corrected each period, suggesting fast convergence in response to spending shocks. This serves as an important validation of the model, as it confirms that deviations from equilibrium are not persistent.

Despite the smaller long-run coefficient for tertiary education, the slightly faster adjustment speed suggests a tighter and more stable long-term relationship with output. This may reflect a higher degree of specialization and direct productivity effects associated with higher education, even if the returns take longer to accumulate or affect a narrower segment of the population.

Interestingly, while the long-term coefficient for secondary education is comparable to that of primary, its short-term effect is slightly larger. This suggests that investment at this level may yield more immediate gains, possibly due to improved workforce readiness or reductions in dropout rates.

The full-model R^2 values exceed 0.83 across all specifications (Table 3). It is important to note that these values reflect the explanatory power of the entire heterogeneous specification, which includes country-specific intercepts (Fixed-Effects), country-specific slope coefficients, lagged dependent variables, and cross-sectional averages—amounting to over 200 estimated parameters. High R^2 values are therefore expected by construction: GDP per capita is highly autoregressive, and the inclusion

⁸It should be noted that the elasticity of aggregate expenditure growth might not be equivalent to the sum of its components (primary, secondary, and tertiary), as all variables are in logarithmic scale.

of its lag alone accounts for a substantial share of within-country variation. When goodness of fit is evaluated using the Mean Group (MG) estimates—the average coefficients reported and interpreted in this paper—the R^2 is approximately 0.20, a level typical of growth regressions. Both measures are reported in [Table 3](#).

Regarding potential endogeneity, the CS-ARDL framework provides several safeguards. First, although the model includes contemporaneous regressors, the dynamic structure ensures that identification relies primarily on lagged variation in expenditure, mitigating concerns about simultaneous determination with current output. Second, country fixed effects absorb all time-invariant unobserved heterogeneity, while cross-sectional averages capture time-varying common factors ([Pesaran, 2006](#)). Third, [Chudik et al \(2016\)](#) show that the CS-ARDL estimator remains consistent even when regressors are only weakly exogenous, which is the realistic scenario for the relationship between public spending and economic growth. While we cannot fully rule out reverse causality at the contemporaneous level, the error correction structure focuses on long-run equilibrium relationships, which are less susceptible to short-run simultaneity bias.

The last rows of [Table 3](#) report the results of the cross-sectional dependence (CD) and heterogeneous slopes (HS) tests. The CD test, based on the null hypothesis of no cross-sectional dependence, yields p-values well above conventional significance levels, indicating that the residuals are not correlated across countries. This suggests that the estimation strategy, which includes cross-sectional averages to capture unobserved common factors, is sufficient to account for any potential cross-sectional dependence. The HS test strongly rejects the null hypothesis of slope homogeneity (p-value = 0.00), indicating that the effect of education expenditure on growth varies meaningfully across countries. However, this heterogeneity is explicitly accommodated in the CS-ARDL framework through the use of country-specific coefficients and by averaging long-run effects. As a result, the estimates remain consistent and robust, without relying on potentially biased pooled estimates ([Ditzen, 2021](#)). The model’s structure is therefore well-suited for this context, and the test results validate the appropriateness of the methodology chosen.

Given that the sample spans nearly five decades (1970–2019), a period marked by several global economic disruptions—including oil price shocks, the Latin American debt crisis, the Asian financial crisis, and the 2008 global financial crisis—we assess the structural stability of the estimated relationships. We apply the panel structural break tests of [Ditzen et al \(2025\)](#), which extend the Bai–Perron framework to panel data with cross-sectional dependence. The tests consider three complementary hypotheses: (i) no break versus a fixed number of breaks; (ii) no break versus an unknown number of breaks (weighted double maximum test); and (iii) a sequential test of s versus $s + 1$ breaks. All tests are conducted with cross-sectional averages and HAC-robust variance estimation, consistent with the CS-ARDL framework. As reported in [Table E5](#) in the Appendix, the test statistics are well below conventional critical values across all specifications and education levels, providing no evidence of structural instability in the coefficient of education expenditure. This result reinforces confidence in the stability of the estimated long-run relationships and is consistent with the CS-ARDL

estimator's inclusion of cross-sectional averages, which absorb common shocks and mitigate the impact of global crises on the estimated coefficients.

In summary, this analysis reinforces the finding that public education spending is a significant driver of economic growth, with notable effects observed across all levels of education. Both the magnitude and timing of these effects vary by educational level, suggesting that a balanced and coordinated investment strategy, targeting both foundational and higher education, may be the most effective approach to foster long-term development.

Table 3: Short- and Long-Term Coefficients of CS-ARDL(p,q) by Educational Levels

Δ Real GDP pc (log)	Aggregate	Primary	Secondary	Tertiary
Period	(1970-2019)	(1970-2019)	(1970-2019)	(1970-2019)
$(p^*, q^*)^\dagger$	(1,0) §	(1,0) §	(1,0) §	(1,0) §
Long term				
Real gov. education exp. pc (log)	0.031*** (0.010)	0.019** (0.009)	0.019** (0.008)	0.013* (0.006)
Short term				
Δ Real gov. education exp. pc (log)	0.024*** (0.007)	0.014** (0.006)	0.015** (0.006)	0.010** (0.005)
Δ Real GDP pc (log) $_{t-1}$	0.226*** (0.024)	0.233*** (0.024)	0.230*** (0.024)	0.222*** (0.024)
Adjustment term				
$-\alpha$	-0.774*** (0.024)	-0.767*** (0.024)	-0.770*** (0.024)	-0.778*** (0.024)
Countries	107	107	107	107
Observations	4,141	4,208	4,196	4,208
Avg. obs. per country	40	41	41	41
R^2 (full model)	0.84	0.84	0.84	0.83
R^2 (MG estimates)	0.20	0.20	0.20	0.20
Tests				
CD p-value	0.42	0.19	0.63	0.93
HS p-value	0.00	0.00	0.00	0.00

Note: † Optimal lags p and q are determined for the full panel using the Bayesian information criterion (BIC). § lags p and q apply to the full panel. All expenditure levels are expressed in real terms, per capita and in logarithms. The dependent variable is the logarithm of real output per capita. The estimates are based on the same set of countries that contain available data for all educational levels. Aggregate expenditure = Primary expenditure + Secondary expenditure + Tertiary expenditure. R^2 (full model) is the goodness of fit of the complete heterogeneous specification, which includes country-specific intercepts, country-specific slopes, and cross-sectional averages (over 200 parameters). R^2 (MG estimates) is computed using only the Mean Group coefficients reported in the table, and provides a more informative measure of the explanatory power of the average education-output relationship. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

5 Country Heterogeneity

In this section, we examine whether the elasticity of output with respect to education expenditure varies across different groups of countries. Of particular interest is the potential role of the quality of expenditure in shaping this relationship. Education

quality serves as a proxy for productivity in the education sector, which may translate into improved labor market outcomes and, ultimately, higher aggregate output. This analysis seeks to explore the mechanisms that could explain cross-country variation in long-term elasticities. Specifically, we investigate whether higher educational achievements, as a proxy for quality, enhance the effectiveness of education spending. If both the expenditure and quality jointly contribute to stronger outcomes, this would provide suggestive evidence that their interaction may be central to economic development. It is important to note that we acknowledge that PISA results may be influenced by the level of per-capita expenditure itself and recognize the complexity of non-linearities present in this exercise.⁹

To implement this, we retrieve the vector of long-term coefficients for each country and conduct cross-country regressions using country-specific characteristics. To account for regional differences, we include a set of region-specific dummies in the regression. Let θ_i denote the country-specific long-run coefficient from the CS-ARDL estimation. We estimate the following specification:

$$\theta_i = \beta + \gamma \text{Math}_i + \mathbf{D}_i' \boldsymbol{\delta} + \epsilon_i, \quad (3)$$

where Math_i represents the logarithm of the initial PISA math score for country i ; $\mathbf{D}_i$ is a vector of regional dummy variables for country i ; and $\boldsymbol{\delta}$ captures their associated coefficients.

The results presented in [Table 4](#) indicate that higher average PISA math scores are associated with larger long-run elasticities of output with respect to education expenditure.¹⁰ This suggests that, on average, countries with better education quality experience a stronger impact of public education spending on long-term economic growth. The relationship is statistically significant both in the aggregate and across all education levels (primary, secondary, and tertiary).

For instance, the coefficient of 0.320 in the aggregate expenditure regression implies that a 10 percent increase in the average historical PISA math score is associated with a 0.032 increase in the long-run elasticity of output with respect to education expenditure, on average. Since the PISA score is included in the regression in logarithmic form, these estimates can be interpreted as semi-elasticities, allowing for a straightforward interpretation of proportional changes.

When comparing across educational levels, we observe that the long-run elasticity associated with primary education (0.283) is slightly higher than those for secondary (0.237) and tertiary education (0.217). However, all effects are statistically significant and economically meaningful. This pattern may reflect the broader reach and foundational role of primary education in developing basic skills that support long-term productivity. Nonetheless, the relatively strong effects at the secondary level, coupled with its sensitivity to education quality, highlight its importance as a critical stage for

⁹As noted in [Table C3](#), countries with higher PISA scores also exhibit higher per capita expenditure on education.

¹⁰Comparable results are obtained when using PISA Reading or Science scores rather than math; see [Table F6](#) and [Table F7](#) in the Appendix.

maximizing returns on investment, particularly in countries seeking to build a more skilled workforce.

Importantly, the relationship between PISA scores and short-term elasticities is weaker and statistically insignificant. In fact, the estimated short-term coefficients are close to zero and occasionally negative, suggesting that improvements in education quality do not yield immediate output gains, but rather accumulate their effects over time.

Regional control variables reveal notable patterns. For long-term elasticities, the dummy variables for Latin America and the Caribbean, the Middle East and North Africa, and Europe and Central Asia, all load positively and significantly in most specifications, relative to the omitted reference category: East Asia and the Pacific. This indicates potential region-specific differences in the education-output relationship that may reflect variations in institutional, demographic, or policy environments.

Table 4: Returns Based on PISA Mathematics Performance

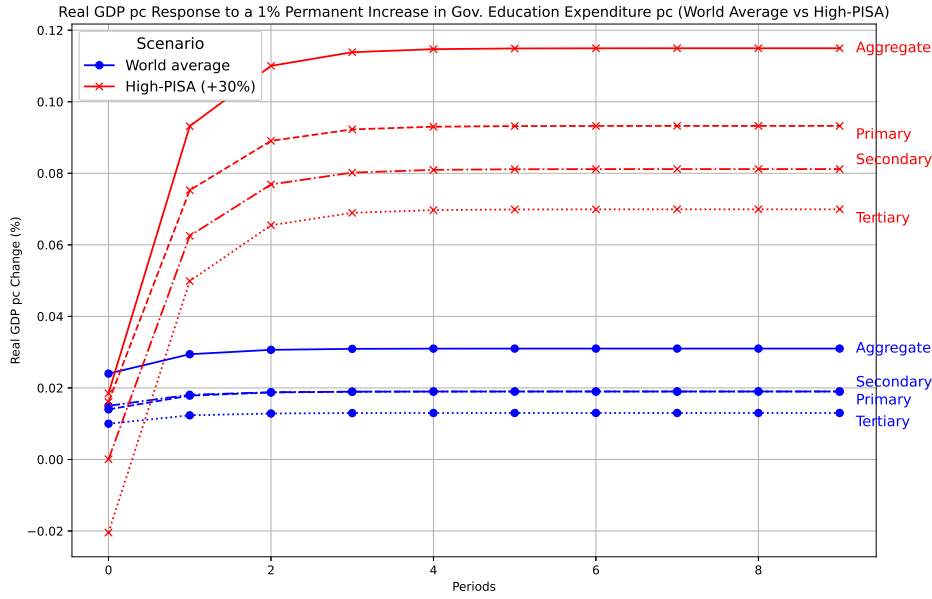
Period	Aggregate expenditure (1970-2019)	Primary (1970-2019)	Secondary (1970-2019)	Tertiary (1970-2019)
<i>Panel A: Long-term education expenditure</i>				
Average PISA math score (log)	0.320** (0.127)	0.283** (0.111)	0.237** (0.108)	0.217** (0.084)
Europe and Central Asia	0.048 (0.038)	0.006 (0.033)	0.021 (0.032)	0.041 (0.025)
Latin America and Caribbean	0.141** (0.054)	0.111** (0.047)	0.078* (0.046)	0.090** (0.036)
Middle East and North Africa	0.162*** (0.058)	0.169*** (0.050)	0.142*** (0.049)	0.081** (0.038)
Constant	-2.008** (0.786)	-1.771** (0.685)	-1.487** (0.666)	-1.360** (0.517)
Countries	54	54	54	54
R-squared	0.18	0.22	0.17	0.18
<i>Panel B: Short-term education expenditure</i>				
Average PISA math score (log)	-0.021 (0.547)	0.008 (0.549)	-0.057 (0.542)	-0.116 (0.531)
Europe and Central Asia	0.231 (0.164)	0.224 (0.165)	0.215 (0.163)	0.253 (0.159)
Latin America and Caribbean	0.255 (0.234)	0.263 (0.235)	0.246 (0.232)	0.233 (0.227)
Middle East and North Africa	0.102 (0.247)	0.088 (0.248)	0.102 (0.245)	0.140 (0.240)
Constant	-0.232 (3.374)	-0.410 (3.389)	0.000 (3.342)	0.327 (3.273)
Countries	54	54	54	54
R-squared	0.05	0.05	0.05	0.06

Note: The dependent variable is the country-specific long-run coefficient (Panel A) or short-run coefficient (Panel B) approximating the elasticity of output with respect to education expenditure. The regression uses the earliest available average PISA math score as the independent variable. Countries listed in Table B2 that lack PISA data are excluded from the analysis. Full-sample estimates are performed on the same set of countries with available information for all education levels. Aggregate expenditure = Primary expenditure + Secondary expenditure + Tertiary expenditure. The East Asia and Pacific fixed effect is omitted due to collinearity. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

6 Dynamics Based on Educational Achievement

This section examines how improvements in education quality, proxied by PISA math scores, affect the macroeconomic impact of public education expenditure. We simulate the dynamic response of real GDP per capita to a 1 percent *permanent* increase in real per capita government education expenditure under two scenarios: (1) a baseline scenario reflecting world average elasticities, and (2) a scenario calibrated for countries with PISA math scores approximately 30 percent higher, corresponding to the observed gap between emerging and advanced economies as documented in [Table C3](#).

Fig. 2: Simulated Response of Real GDP Per Capita (percent) to a 1 percent Permanent Increase in Real Government Education Expenditure Per Capita



Note: Simulated cumulative response of real GDP per capita to a 1 percent permanent increase in real per capita government education expenditure. The baseline scenario (World Average) uses the Mean Group long-run elasticities from [Table 3](#). The high-PISA scenario calibrates the elasticities for countries with PISA math scores approximately 30 percent higher, corresponding to the gap between emerging and advanced economies, using the coefficients from [Table 4](#). Panels correspond to aggregate, primary, secondary, and tertiary education expenditure.

Figure 2 illustrates that higher educational achievement substantially amplifies the long-run impact of public education spending on GDP. In the aggregate case, the cumulative effect on GDP per capita more than triples, from approximately 0.031 percent under the world average to approximately 0.115 percent in the high-PISA scenario. This underscores the catalytic role of education quality in enhancing the productivity of public investment.

When disaggregated by education level, primary and secondary education display particularly pronounced responses to improved achievement. Under high-PISA performance, the increases in GDP per capita peak at around 0.093 percent and 0.081

percent, respectively, roughly four times the effects observed under average conditions (0.019 percent each). These results suggest that foundational and intermediate stages of education are especially sensitive to quality improvements, likely reflecting systemic gains in instructional delivery and student learning outcomes.

Tertiary education, while starting from a smaller baseline, shows the largest relative gain, with GDP effects rising from roughly 0.013 percent to approximately 0.070 percent, more than a fivefold increase. This substantial jump highlights that even advanced levels of education can yield significant macroeconomic returns when quality improves.

Across all education levels, the effects stabilize within a few periods, reflecting the time required for human capital investments to translate into productivity gains. Nevertheless, these gains are persistent, reinforcing the case for sustained quality-focused education policy.

In sum, these findings suggest that improvements in education quality, through better teaching, curricula, or student outcomes, can significantly enhance long-run macroeconomic returns on public education spending. For policymakers, this underscores the importance of complementing increased expenditure with reforms aimed at improving system effectiveness and learning quality.

7 Discussion

This paper provides new evidence on the dynamic relationship between public education expenditure and economic growth using a large global panel and a unified short- and long-run framework. The results yield several important insights that help reconcile prior mixed findings in the literature and clarify the conditions under which education spending contributes to macroeconomic performance. In this section, we interpret the main findings, relate them to existing empirical and theoretical work, discuss plausible mechanisms, and draw policy implications, while acknowledging the study's limitations.

A central result of the paper is that public education spending exhibits statistically significant and economically meaningful effects on output in both the short and long run, with long-run elasticities substantially larger than short-run responses. The estimated long-run elasticity of approximately 0.03 for aggregate education expenditure implies that sustained increases in per capita education spending translate into persistent gains in GDP per capita. While the magnitude may appear modest in annual terms, it is sizable when interpreted as a level effect that compounds over time through human capital accumulation.

This dynamic pattern aligns closely with the intuition emphasized in the endogenous growth literature, where education affects productivity gradually through skill formation and knowledge diffusion (Lucas, 1988; Barro, 1990). Empirically, the results are consistent with ARDL- and VECM-based studies that document weak or modest short-run effects but robust long-run impacts of education spending on output (Afzal et al, 2010; Adeyemi and Ogunsola, 2016; Ifa and Guetat, 2018; Mallick et al, 2016). By explicitly estimating both horizons in a global panel, the paper helps reconcile earlier cross-country studies that failed to detect significant effects, often because

they implicitly conflated short-run dynamics with long-run equilibria (Landau, 1986; Levine and Renelt, 1992).

At the same time, the presence of positive and statistically significant short-run effects suggests that education spending can stimulate economic activity through channels other than human capital accumulation alone. These may include demand-side mechanisms such as employment in the education sector, procurement of goods and services, and infrastructure investment, echoing findings from fiscal multiplier studies focused on education expenditure (Angelopoulos et al, 2008; De Ridder et al, 2020). The coexistence of short-run demand effects and long-run supply-side gains underscores the multifaceted role of education spending in macroeconomic dynamics.

Disaggregating education expenditure by level reveals important heterogeneity in growth returns. Primary and secondary education expenditures exhibit larger long-run elasticities than tertiary education, although all levels contribute positively to output. This pattern is broadly consistent with the microeconomic literature on social returns to education, which finds particularly high returns to foundational and intermediate schooling, especially in developing and middle-income contexts (Psacharopoulos and Patrinos, 2018).

From a macroeconomic perspective, stronger effects at the primary and secondary levels may reflect their broader coverage and their role in building basic and intermediate skills that raise productivity across a large share of the labor force. These findings are in line with evidence from country-level and regional studies showing that investments in basic education are closely linked to growth through labor force participation, adaptability, and reductions in skill mismatches (Self and Grabowski, 2004; Kiran, 2014). In contrast, tertiary education, while essential for innovation and frontier technological progress, directly affects a smaller segment of the population, which may explain its comparatively lower average elasticity in aggregate output.

Importantly, the results do not imply that tertiary education is unimportant for growth. Rather, they suggest that its macroeconomic impact may depend more strongly on complementarities with labor market conditions, innovation systems, and absorptive capacity, factors that vary widely across countries and are not explicitly modeled here. The relatively fast adjustment speed estimated for tertiary education expenditure is consistent with a tighter long-run relationship once these complementarities are in place.

One of the most consequential findings of the paper is that the growth impact of education spending is strongly conditioned by education quality, proxied by PISA test scores. Countries with higher learning outcomes experience substantially larger long-run elasticities of output with respect to education expenditure, while short-run effects remain largely insensitive to quality differences.

This result provides macroeconomic support for the argument advanced by Hanushek and Kimko (2000) and Hanushek and Woessmann (2010), who emphasize that cognitive skills, rather than years of schooling or spending alone, are the key driver of growth. The findings also help explain why earlier studies reported weak or insignificant effects of education spending: if additional resources do not translate

into improved learning, their long-run productivity impact is limited (Pritchett, 2001; Blankenau and Simpson, 2004).

The simulations further highlight the nonlinear nature of this relationship. A given increase in education spending yields dramatically larger output gains in high-achievement environments, suggesting that quality acts as a multiplier on investment. This complements evidence from Baldacci et al (2008), who find that social spending promotes growth primarily through improvements in human capital outcomes, and extends it by quantifying how learning quality amplifies the returns to education expenditure in a global setting.

Taken together, the results point to several reinforcing mechanisms. First, higher-quality education systems are likely more effective at converting additional resources into genuine skill acquisition, thereby strengthening the link between spending and productivity. Second, education quality may proxy for broader institutional capacity, such as governance, accountability, and curriculum alignment, that enhances the efficiency of public investment. Third, complementarities between education quality and other growth drivers (technology adoption, labor market flexibility, and innovation systems) may further magnify long-run effects.

The absence of a strong relationship between education quality and short-run effects is also informative. It suggests that demand-side channels operate similarly across countries, while supply-side productivity gains depend critically on learning outcomes. This distinction reinforces the importance of adopting a dynamic framework that separates immediate fiscal effects from long-term growth mechanisms.

The findings carry clear policy implications. First, increasing education spending can contribute to economic growth, but the magnitude of the payoff depends crucially on how effectively resources are transformed into learning. Policies that focus solely on expanding budgets without addressing quality may yield limited long-run returns. Second, investments in primary and secondary education appear particularly important for broad-based growth, especially in countries seeking to raise average productivity and reduce inequality. Third, improving education quality through teacher effectiveness, curriculum reform, assessment systems, and accountability, can substantially enhance the growth dividends of education spending.

From a fiscal perspective, the results suggest that education spending should be evaluated not only in terms of its size but also in terms of its efficiency and complementarities with institutional reforms. This perspective aligns with the view that education policy is most effective when integrated into a broader development strategy that strengthens state capacity and labor market institutions.

8 Conclusion

This paper asked whether public education expenditure has measurable effects on economic growth in the short and long run, and how these effects vary by education level and by education quality. Consistent with our stated objective and supporting our hypothesis, we have explored this relationship using a large global panel and a

CS-ARDL framework. Our findings show that investment in education is a significant driver of real GDP per capita, providing robust evidence of both short- and long-term effects. While short-run responses are modest, long-run elasticities are positive and statistically significant across all educational levels, most notably for aggregate spending, but also for primary and secondary education individually.

Importantly, we find strong support for cointegration between education spending and output, thereby validating the use of an error correction specification. The model demonstrates rapid convergence toward the long-run equilibrium, with about 77 percent of deviations corrected each period. This finding suggests that education investment not only yields persistent effects but also contributes to macroeconomic stability.

A key contribution of this study lies in identifying heterogeneity in the effectiveness of education spending. We find that countries with higher educational achievement, as proxied by PISA math scores, experience significantly greater returns on investment. This effect is particularly notable in primary and secondary education, where quality improvements appear to most strongly enhance productivity. Conversely, short-term effects are not significantly linked to educational quality, reinforcing the notion that gains from improved education systems accumulate over time.

The simulations further illustrate how education quality acts as a multiplier: when PISA scores increase by 30 percent, roughly the gap between emerging and advanced economies, the long-term impact of education expenditure on GDP per capita more than triples. This suggests that the marginal productivity of spending is not uniform across countries but is conditional on the system's performance.

In sum, these results underscore the importance of both the quantity and quality of public education investment. While increasing education budgets is beneficial, the growth dividends are substantially higher when paired with improvements in learning outcomes. Policymakers should therefore focus not only on expanding access and funding, but also on systemic reforms aimed at enhancing instructional quality, curriculum alignment, and educational accountability.

Several limitations should be noted. First, the analysis relies on country-level data; subnational variation in spending and outcomes is not captured. Second, PISA coverage varies across countries and years, which may affect the heterogeneity estimates. Third, although panel structural break tests indicate no evidence of instability in the education-output coefficient over the sample period, the analysis does not model regime-specific coefficients, so the average effects may not fully capture time variation during crisis episodes. Fourth, although the CS-ARDL framework accounts for cross-sectional dependence and heterogeneity, reverse causality between growth and education spending cannot be fully ruled out.

Future research could extend this work in several directions. Subnational data would allow testing whether the aggregate relationships hold within countries. Incorporating additional measures of education quality or spending efficiency could sharpen the heterogeneity analysis. Finally, combining expenditure data with detailed policy

indicators could help identify which types of reforms most enhance the productivity of education spending.

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Appendices

A Variables and Sources

Table A1: Variable Description and Sources

Variable	Description	Source
GDP	GDP in current US dollars (millions)	World Bank (NY.GDP.MKTP.CD)
Y	Real GDP in constant 2015 US dollars (millions)	World Bank (NY.GDP.MKTP.KD)
Deflator	GDP deflator (base year 2015), calculated as $GDP / Y \times 100$	Constructed from World Bank (NY.GDP.MKTP.CD / NY.GDP.MKTP.KD)
Population	Total population by country and year	World Bank (SP.POP.TOTL)
Real gov. education exp. pc (log)	Log of total real government education expenditure per capita (sum of primary, secondary, and tertiary nominal expenditures deflated and divided by population)	Constructed from UIS + World Bank
Real gov. primary education exp. pc (log)	Log of real government primary education expenditure per capita (nominal primary expenditure deflated and divided by population)	Constructed from UIS.X.US.1.FSGOV + World Bank
Real gov. secondary education exp. pc (log)	Log of real government secondary education expenditure per capita (nominal secondary expenditure deflated and divided by population)	Constructed from UIS.X.US.2T3.FSGOV + World Bank
Real gov. tertiary education exp. pc (log)	Log of real government tertiary education expenditure per capita (nominal tertiary expenditure deflated and divided by population)	Constructed from UIS.X.US.5T8.FSGOV + World Bank
Average PISA math score	Average PISA mathematics score by country and year	OECD PISA Dataset

Note: Description of all variables used in the analysis, including their construction and original data sources. GDP and population data are from the World Bank's World Development Indicators (WDI). Education expenditure data by level are from UNESCO's Institute for Statistics (UIS). All monetary variables are deflated using the GDP deflator (base year 2015) and expressed in per capita terms.

B Countries in Sample

Table B2: List of Countries Included in the Sample

<i>Advanced (35)</i>	<i>Emerging and Developing (72)</i>		
Australia	Albania	Algeria	Argentina
Austria	Bahrain	Bangladesh	Barbados
Belgium	Bermuda	Bhutan	Botswana
Chile	Brazil	Brunei Darussalam	Burkina Faso
Cyprus	Burundi	Cameroon	Central African Republic
Czech Republic	Colombia	Comoros	Congo, Rep
Denmark	Costa Rica	Cote d'Ivoire	Dominica
Finland	Dominican Republic	Ecuador	El Salvador
France	Eswatini	Ethiopia	Fiji
Germany	Gabon	Gambia, The	Ghana
Greece	Guatemala	Guinea	Guyana
Hong Kong SAR, China	Honduras	Indonesia	Iran, Islamic Rep
Hungary	Jamaica	Jordan	Kenya
Iceland	Kuwait	Lao PDR	Lesotho
Ireland	Liberia	Madagascar	Malawi
Israel	Malaysia	Mali	Mauritania
Italy	Mauritius	Mexico	Morocco
Japan	Nicaragua	Niger	Oman
Korea, Rep	Pakistan	Paraguay	Peru
Luxembourg	Philippines	Rwanda	Seychelles
Malta	Sri Lanka	St Kitts and Nevis	St Lucia
Netherlands	Syrian Arab Republic	Tanzania	Thailand
New Zealand	Togo	Tonga	Tunisia
Norway	Uganda	Zambia	Zimbabwe
Panama			
Portugal			
Puerto Rico			
Singapore			
Slovak Republic			
Slovenia			
Spain			
Sweden			
Switzerland			
United Kingdom			
Uruguay			

Note: The 107 countries in the sample, classified as Advanced (World Bank high-income economies, excluding oil-rich states and small island economies) or Emerging and Developing (all others). The sample is determined by the availability of both GDP data (World Bank, WDI) and government education expenditure data by level (UNESCO, UIS) over the period 1970–2019.

C Descriptive Statistics

Table C3: Descriptive Statistics for the Different Samples

	Mean	Std. Dev.	Min.	Max.
<i>Whole Sample (107)</i>				
Real GDP pc (log)	8.444	1.583	-6.057	11.795
Real gov. education exp. pc (log)	5.173	1.741	0.227	8.487
Real gov. primary education exp. pc (log)	4.264	1.603	-0.846	7.710
Real gov. secondary education exp. pc (log)	4.112	1.921	-1.777	7.627
Real gov. tertiary education exp. pc (log)	3.516	1.950	-7.380	7.429
PISA math score	447.114	75.689	292.070	563.816
<i>Advanced (35)</i>				
Real GDP pc (log)	9.964	0.761	6.935	11.630
Real gov. education exp. pc (log)	6.811	0.903	4.041	8.487
Real gov. primary education exp. pc (log)	5.703	0.873	2.838	7.710
Real gov. secondary education exp. pc (log)	5.903	0.994	2.651	7.627
Real gov. tertiary education exp. pc (log)	5.293	1.093	1.503	7.429
PISA math score	496.438	48.717	359.748	563.816
<i>Emerging and Developing (72)</i>				
Real GDP pc (log)	7.698	1.328	-6.057	11.795
Real gov. education exp. pc (log)	4.311	1.429	0.227	7.860
Real gov. primary education exp. pc (log)	3.517	1.370	-0.846	7.151
Real gov. secondary education exp. pc (log)	3.192	1.608	-1.777	7.170
Real gov. tertiary education exp. pc (log)	2.566	1.614	-7.380	6.830
PISA math score	369.605	34.199	292.070	432.303

Note: Summary statistics (mean, standard deviation, minimum, maximum) for the main variables, reported separately for the whole sample and by income group. All variables are in real per capita terms and in logarithms. Data sources: World Bank (WDI) for GDP, UNESCO (UIS) for education expenditure, and OECD for PISA scores.

D Correlations

Table D4: Correlation Between Variables of Interest for Different Samples

	Real GDP pc (log)	Real gov. education exp. pc (log)	Real gov. primary education exp. pc (log)	Real gov. secondary education exp. pc (log)	Real gov. tertiary education exp. pc (log)
<i>Whole Sample (107)</i>					
Real GDP pc (log)	1				
Real gov. education exp. pc (log)	0.961***	1			
Real gov. primary education exp. pc (log)	0.945***	0.983***	1		
Real gov. secondary education exp. pc (log)	0.946***	0.984***	0.950***	1	
Real gov. tertiary education exp. pc (log)	0.901***	0.946***	0.909***	0.915***	1
<i>Advanced (35)</i>					
Real GDP pc (log)	1				
Real gov. education exp. pc (log)	0.940***	1			
Real gov. primary education exp. pc (log)	0.873***	0.933***	1		
Real gov. secondary education exp. pc (log)	0.920***	0.963***	0.844***	1	
Real gov. tertiary education exp. pc (log)	0.784***	0.865***	0.739***	0.796***	1
<i>Emerging and Developing (72)</i>					
Real GDP pc (log)	1				
Real gov. education exp. pc (log)	0.925***	1			
Real gov. primary education exp. pc (log)	0.910***	0.980***	1		
Real gov. secondary education exp. pc (log)	0.898***	0.972***	0.927***	1	
Real gov. tertiary education exp. pc (log)	0.824***	0.909***	0.862***	0.857***	1

Note: Pairwise correlations between the main variables, reported separately for the whole sample and by income group. All variables are in real per capita terms and in logarithms. Data sources: World Bank (WDI) for GDP, UNESCO (UIS) for education expenditure, and OECD for PISA scores. Significance: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

E Structural Break Tests

Table E5: Panel Structural Break Tests

	Aggregate	Primary	Secondary	Tertiary
Panel A: H_0: no break vs H_1: s breaks				
$s = 1$	0.352 [7.04, 8.58, 12.29]	0.846 [7.04, 8.58, 12.29]	0.824 [7.04, 8.58, 12.29]	1.251 [7.04, 8.58, 12.29]
$s = 2$	1.123 [6.28, 7.22, 9.36]	0.592 [6.28, 7.22, 9.36]	1.132 [6.28, 7.22, 9.36]	3.116 [6.28, 7.22, 9.36]
$s = 3$	1.540 [5.21, 5.96, 7.60]	0.672 [5.21, 5.96, 7.60]	1.301 [5.21, 5.96, 7.60]	2.019 [5.21, 5.96, 7.60]
Panel B: H_0: no break vs H_1: $1 \leq s \leq 3$ breaks (WDmax)				
WDmax	2.218 [8.20, 9.91, 13.83]	0.967 [8.20, 9.91, 13.83]	1.872 [8.20, 9.91, 13.83]	3.703 [8.20, 9.91, 13.83]
Panel C: H_0: s breaks vs H_1: $s + 1$ breaks (sequential)				
$F(2 1)$	5.745 [8.51, 10.13, 13.89]	3.099 [8.51, 10.13, 13.89]	4.317 [8.51, 10.13, 13.89]	5.529 [8.51, 10.13, 13.89]
$F(3 2)$	3.213 [9.41, 11.14, 14.80]	2.940 [9.41, 11.14, 14.80]	0.490 [9.41, 11.14, 14.80]	0.265 [9.41, 11.14, 14.80]

Note: Structural break tests based on Ditzén, Karavias & Westerlund (2025). The dependent variable is the first difference of log real GDP per capita. All specifications include country fixed effects, cross-sectional averages to account for cross-sectional dependence, and HAC-robust standard errors. Panel A tests the null of no structural break against the alternative of exactly s breaks ($s = 1, 2, 3$). Panel B reports the weighted double maximum (WDmax) test for the null of no break against an unknown number of breaks ($1 \leq s \leq 3$). Panel C reports the sequential test of s vs. $s + 1$ breaks. F-statistics are reported with critical values at the 10%, 5%, and 1% significance levels in brackets. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

F Robustness: PISA Reading and Science

Table F6: Returns Based on PISA Reading Performance

Period	Aggregate expenditure (1970-2019)	Primary (1970-2019)	Secondary (1970-2019)	Tertiary (1970-2019)
<i>Panel A: Long-term education expenditure</i>				
Average PISA reading score (log)	0.368*** (0.131)	0.325*** (0.114)	0.256** (0.112)	0.254*** (0.086)
Europe and Central Asia	0.035 (0.038)	-0.005 (0.033)	0.013 (0.033)	0.032 (0.025)
Latin America and Caribbean	0.112** (0.046)	0.086** (0.040)	0.054 (0.040)	0.071** (0.030)
Middle East and North Africa	0.157*** (0.055)	0.164*** (0.048)	0.136*** (0.047)	0.078** (0.036)
Constant	-2.287*** (0.805)	-2.020*** (0.701)	-1.590** (0.689)	-1.577*** (0.527)
Countries	54	54	54	54
R-squared	0.20	0.24	0.18	0.21
<i>Panel B: Short-term education expenditure</i>				
Average PISA reading score (log)	0.186 (0.570)	0.219 (0.573)	0.115 (0.566)	0.052 (0.554)
Europe and Central Asia	0.218 (0.167)	0.210 (0.168)	0.205 (0.166)	0.245 (0.162)
Latin America and Caribbean	0.291 (0.201)	0.296 (0.202)	0.280 (0.199)	0.273 (0.195)
Middle East and North Africa	0.133 (0.241)	0.119 (0.242)	0.130 (0.238)	0.169 (0.234)
Constant	-1.499 (3.498)	-1.706 (3.512)	-1.061 (3.468)	-0.704 (3.399)
Countries	54	54	54	54
R-squared	0.05	0.06	0.05	0.06

Note: The dependent variable is the country-specific long-run coefficient (Panel A) or short-run coefficient (Panel B) approximating the elasticity of output with respect to education expenditure. The regression uses the earliest available average PISA reading score as the independent variable. Countries listed in [Table B2](#) that lack PISA data are excluded from the analysis. Full-sample estimates are performed on the same set of countries with available information for all education levels. Aggregate expenditure = Primary expenditure + Secondary expenditure + Tertiary expenditure. The East Asia and Pacific fixed effect is omitted due to collinearity. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table F7: Returns Based on PISA Science Performance

	Aggregate expenditure	Primary	Secondary	Tertiary
Period	(1970-2019)	(1970-2019)	(1970-2019)	(1970-2019)
Panel A: Long-term education expenditure				
Average PISA science score (log)	0.382** (0.145)	0.337** (0.126)	0.269** (0.123)	0.251** (0.096)
Europe and Central Asia	0.047 (0.038)	0.005 (0.033)	0.021 (0.032)	0.041 (0.025)
Latin America and Caribbean	0.132** (0.051)	0.103** (0.045)	0.068 (0.044)	0.083** (0.034)
Middle East and North Africa	0.160*** (0.057)	0.167*** (0.049)	0.139*** (0.048)	0.079** (0.037)
Constant	-2.387** (0.893)	-2.105*** (0.777)	-1.679** (0.761)	-1.573** (0.589)
Countries	54	54	54	54
R-squared	0.19	0.23	0.17	0.18
Panel B: Short-term education expenditure				
Average PISA science score (log)	-0.017 (0.625)	0.024 (0.628)	-0.079 (0.619)	-0.106 (0.606)
Europe and Central Asia	0.231 (0.164)	0.224 (0.165)	0.215 (0.163)	0.252 (0.159)
Latin America and Caribbean	0.257 (0.221)	0.265 (0.222)	0.245 (0.219)	0.243 (0.214)
Middle East and North Africa	0.103 (0.244)	0.090 (0.245)	0.101 (0.242)	0.146 (0.237)
Constant	-0.255 (3.853)	-0.510 (3.870)	0.136 (3.816)	0.267 (3.739)
Countries	54	54	54	54
R-squared	0.05	0.05	0.05	0.06

Note: The dependent variable is the country-specific long-run coefficient (Panel A) or short-run coefficient (Panel B) approximating the elasticity of output with respect to education expenditure. The regression uses the earliest available average PISA science score as the independent variable. Countries listed in Table B2 that lack PISA data are excluded from the analysis. Full-sample estimates are performed on the same set of countries with available information for all education levels. Aggregate expenditure = Primary expenditure + Secondary expenditure + Tertiary expenditure. The East Asia and Pacific fixed effect is omitted due to collinearity. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.